FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 06-06-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABCD

Sol. The potential inside the outer shell is same throughout hence (A) The p d between the shell decrease hence (B)

Earthing always is zero potential hence (C)

If 'x' is the change on inner sphere then $\frac{Kx}{R}+\frac{KQ}{2R}=0$

$\therefore x=-\frac{Q}{2}$ Hence (D)

2. BCD

Sol. Required minimum frication coefficient for pure-rolling.

$$\mu=\frac{\tan\left(K^2/R^2\right)}{\left(1+K^2/R^2\right)}$$

$$\mu_{hollow\,sphere}=\frac{(3/4)\times 2/3}{(1+2/3)}=\frac{2/4}{5/3}=\frac{2}{3}\times\frac{3}{5}=\frac{3}{10}=0.3$$

$$\mu_{solid\,sphere}=\frac{3/4\times 2/5}{(1+2/5)}=\frac{3/4\times 2/5}{7/5}=\frac{3}{4}\times\frac{2}{5}\times\frac{5}{7}=\frac{3}{14}<\mu_{Hollow\,sphere}$$

$$similarly\ for\ solid\ sphere=\frac{K_R}{K_T}=\frac{2}{5}$$

3. ABD

4. AB

5. BD

6. BC

SECTION – B

7. 4

Sol. For the element

$$i=\frac{dV}{\frac{1}{\sigma}.\frac{dx}{4\pi^2}}=\frac{Edx}{\frac{1}{KE}.\frac{dx}{4\pi^2}}$$

$$i=KE^2 4\pi x^2$$

$$\Rightarrow\sqrt{\frac{i}{4\pi K}}=E.x=\frac{dV}{dx}.x \qquad \Rightarrow\sqrt{\frac{1}{4\pi K}}\int_a^b\frac{dx}{x}=\int_0^V dV$$

$$\Rightarrow i=\frac{4\pi KV^2}{\left(In\frac{b}{a}\right)^2}=\frac{4\pi\times\frac{1}{\pi}\times10^{-2}\times(20)^2}{(1)^2}=16$$

8. 3

Sol. The equation of current in the inductor $i=(6A)\cos t\left(at\,t=0,i\,is\,\max imum\,and\,\omega=1\right)$ and equation of charge on the capacitor

$$q=(6C)\sin t\left(at\,t=0,q=0\right)at\,t=\frac{\pi}{6},q=3C$$

9. 4

10. 4

Sol. $$\frac{n}{2L}\sqrt{\frac{160}{\mu}}=\frac{n-1}{2L}\sqrt{\frac{260}{\mu}}=\frac{1}{2L}\sqrt{\frac{mg}{\mu}}$$

m = maximum mass

$$4n=5n-5$$

$$n=5$$

11. 2

Sol. $$\beta=22\left(\frac{330}{330-10}-\frac{330}{330+10}\right)=33\frac{330(2\times10)}{(330)^2-(10)^2}$$

$$\beta=33\times\frac{20}{330}=2\,Hz$$

12. 6

SECTION – C

13. 00003.00

Sol. $T \propto \sqrt{\text{length}}$ free length becomes $\frac{\ell}{4}$ so time period becomes $T_1 = \frac{T}{2}$

Total time $= \frac{T}{2} + \frac{T_1}{2} = \frac{3T}{4}$

14. 00006.00

Sol. $X_{CM} = \frac{\int dm.x}{\int dm} = \frac{2L}{3}$

$F_B \frac{L}{2}\cos 37° - Mg\frac{2L}{3}\cos 37° = TL\cos 37°$

On solving

$T = \frac{10^4}{6} N$

15. 00008.00

Sol. $2T\cos 37° = Mg$

$3\left(\frac{1}{2d}\sqrt{\frac{T}{\mu}}\right) = 25$

16. 00003.00

Sol. $\frac{1}{f} = \left(\frac{7}{4} - 1\right) \times \frac{1}{6} = \frac{3}{4} \times \frac{1}{6} = \frac{1}{8} \Rightarrow f = 8cm$

For first image due to lens: $\frac{1}{v_1} \times \frac{1}{8} - \frac{1}{24} \times \frac{1}{12} \Rightarrow v_1 = 12cm$

Apparent depth $= 12 \times \frac{4}{3} = 16cm \Rightarrow x = 19 - 16 = 3cm$

17. 00006.00

Sol.

$$\left.\begin{aligned} \frac{1}{m} = \frac{u}{f} - 1 \Rightarrow \frac{1}{m_1} = \frac{25}{20} - 1 \\ \frac{1}{m_2} = \frac{50}{20} - 1 \end{aligned}\right\} \frac{m_1}{m_2} = \frac{50-20}{25-20} = \frac{30}{5} = 6$$

18. 00008.00

Sol. Initial velocity $\vec{u} = 40\hat{i} + 30\hat{j}$

$\vec{a} = -10\hat{j}$

Velocity after 7 second $= 40\hat{i} + (30 - 7 \times 10)\hat{j} = 40\hat{i} - 40\hat{j}$

Weight of the body $= -mg\hat{j} = -0.01 \times 10\hat{j} = -0.1\hat{j}$

Power $= \vec{F}.\vec{v} = 4W$

Chemistry

PART – II

SECTION – A

19. ABCD

20. CD

21. ABC

22. ABC

23. ABCD

24. ABD

SECTION – B

25. 0

Sol. $Ag(CN)_2^- \longrightarrow Ag^+ + 2CN^-$

0.1 x 2x

$4\times10^{-17} = \frac{4x^3}{0.1}$

$x = 10^{-6}$

Ionic product $= \left[Ag^+\right]\left[Cl^-\right] = 10^{-6}\times0.02 = 2\times10^{-8}$

If $K_{sp} <$ Ionic product

So precipitation will take place.

26. 2

Sol. $0.48\ \mu Ci = 3.7\times10^4\times0.48\,\text{dPs}/m\ell$

For $0.5\,m\ell$ drop, it is $\frac{3.7\times10^4\times0.48}{2}dPs$

$\frac{444}{60}\text{dPs} = \frac{3.7\times10^4\times0.24\times0.1}{\text{V}}\text{dPs}$

$V = 120$

So $x = 2$

27. 3

Sol. $E_a = \frac{k_1E_{a_1} + k_2E_{a_2}}{k_1+k_2} = \frac{2E_1+4E_2}{6} = \frac{E_1+2E_2}{3}$

$x = 3$

28. 4

Sol. a,b,d,i,m,,n,o,p,q,

29. 7

Sol. D is wrong other statements are correct

This ore is $FeCr_2O_4$ and it is the prep. Of $K_2Cr_2O_7$ in figure

$FeCr_2O_4+O_2+Na_2CO_3 \rightarrow Na_2CrO_4(\text{yellow})+Fe_2O_3+CO_2$ after acidification we have

$NaCr_2O_7.2H_2O$ and $Na_2SO_4.10H_2O$

30. 9

Sol. $B_3N_3H_6 \xrightarrow{3HCl} B_3N_3H_9Cl_3$

SECTION – C

31. 00009.11

Sol. $Q_{rev.} = nRT\ln\left(\frac{V_2}{V_1}\right) = 1\times8.3\times298\times2.303\times0.477 = 2717.67\ J$

$Q_{rev.} = 2717.67$

$\Delta S_{system} = \frac{Q_{rev.}}{T} = 9.11$

32. 00345.00

Sol. Balance eq. is

$4Au + 8CN^- + O_2 + 2H_2O \rightarrow 4Au(CN)_2^- + 4OH^-$

(i) $4Au \rightarrow 4Au^+ + 4e$ $\quad \Delta G^\circ = 4\times96500\times1.5\ J$

$4Au^+ + 8CN^- \rightarrow 4Au(CN)_2^-$ $\quad \Delta G^\circ = -2\times8\times300\times38\times4\ J$

$4Au + 8CN^- \rightleftharpoons 4Au(CN)_2^- + 4e$ $\quad \Delta G_J^\circ = 579 - 729.6\ KJ = -150.6\ KJ$

(ii) $O_2 + 4H^+ + 4e \rightarrow 2H_2O$ $\quad \Delta G'^\circ = -4\times96500\times1.2\ J$

$4H_2O \rightarrow 4H^+ + 4OH^-$ $\quad \Delta G_2'^\circ = 2\times8\times300\times56$

$O_2 + 2H_2O + 4e \rightarrow 4OH^-$ $\quad \Delta G_3'^\circ = -463.2 + 268.8\ kJ = -194.4\ kJ$

$\Delta G^\circ_{reaction} = \Delta G_3^\circ + \Delta G_3'^\circ = -345\ kJ$

33. 00000.01

Sol. $HCOOH \rightleftharpoons HCOO^- + H^+$

$K_{a_1} = \frac{[HCOO^-][H^+]}{[HCOOH]} = \frac{[H^+]^2}{[HCOOH]}$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$K_{a_2} = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{[H^+]^2}{[CH_3COOH]}$

As $[H^+]$ in both case are equal

$K_{a_1}\times[HCOOH] = K_{a_2}[CH_3COOH]$

$\therefore \quad [CH_3COOH] = \frac{K_{a_1}}{K_{a_2}}\times[HCOOH] = \frac{1.8\times10^{-4}}{1.8\times10^{-5}}\times10^{-3} = 0.01\,M$

34. 00871.10

Sol. $p_{O_2} + p_{H_2O} = 740$

$p_{O_2} = 740 - 31.8$ at $30^\circ C$

= 708.2 torr

$p_{N_2} = 780 - 23.8$ at $25^\circ C$

= 756.2 torr

Mix at $35^\circ C$

$$\frac{p_1V_1}{T_1} + \frac{p_2V_2}{T_2} = \frac{pV}{T}$$

$$\frac{708.2 \times 250}{303} + \frac{756.2 \times 300}{298} = \frac{p \times 500}{308}$$

p = 828.9 torr

Final pressure = p + aqueous tension = 828.9 + 42.2 = 871.1

35. 00020.48

Sol.
$$\lambda_M = \frac{K \times 1000}{M}$$

$$\Rightarrow (80 \times 2 + 40) = \frac{\left(2.00 \times 10^{-6} - 0.40 \times 10^{-6}\right) \times 1000}{M}$$

So, $M = 8 \times 10^{-6}$

$K_{sp} = 2048 \times 10^{-18} = 20.48 \times 10^{-16}$

36. 00013.68

Sol.
$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

14 38.5

0 10.5 14

So 25-24.5=0.5 ml of H_2O vap is present

So, vap. Pressure of water at this temperature= $\frac{0.5}{25} = \frac{1}{50}$ atm

Now apply $\frac{P_A^o - P_{solution}}{P_A^o} = 0.1$ and $P_A^o = \frac{1}{50}$

$\Rightarrow P_{solution} = 13.68 \text{mm of Hg}$

Mathematics

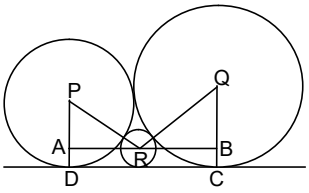
PART – III

SECTION – A

37. AC

38. AD

Sol. Let P and Q be the centres of the circles let $AR = a$ and BR = b and the solution of third circle be r.



$(1-r)^2 + a^2 = (1+r)^2$ and $(2-r)^2 + b^2 = (2+r)^2$

Therefore $a^2 = 4r$ and $b^2 = 8r$

Also $(a+b)^2 + 1 = 9$, so $2\sqrt{r} + 2\sqrt{2r} = \sqrt{8}$

$\therefore \quad r = 6 - 4\sqrt{2}$

39. ACD

Sol. $f(x + y) = g(x) + h(y)$ …(1)

put x = 0

$f(y) = g(0) + h(y) \Rightarrow h(y) = f(y) - g(0)$ …(2)

put y = 0

$f(x) = g(x) + h(0) \Rightarrow g(x) = f(x) - h(0)$ …(3)

$\Rightarrow$ put (2) and (3) in (1)

$f(x + y) = f(y) + f(x) - h(0) - g(0)$

Define $C_1(x) = f(x) - h(0) - g(0) \; \forall \; x \in R$.

$\Rightarrow C_1(x + y) = C_1(x) + C_1(y)$

The solution of this functional. Equation is obtained by differentiation through first principles as $C_1(x) = Cx$ (where 'c' is a constant)

$\Rightarrow f(x) = C_1(x) + h(0) + g(0) = cx + h(0) + g(0)$

$g(x) = f(x) - h(0) = cx + g(0)$

$h(x) = f(x) - g(0) = cx + h(0)$

$f'(0) = f'(1) = c$

40. BCD

41. ABC

Sol. $A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$A^3 - A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ $A - I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

$\Rightarrow A^3 - A^2 = A - I$

$\Rightarrow$ Det $|A^n - I|$ = Det((A – I)(I + A + A^2 + … + A^{n-1}))

= Det (A – I) Det(I + A + A^2 + … + A^{n-1})

= 0

$A^3 - A^2 = A - I$ …(1)

$\Rightarrow A^4 - A^3 = A^2 - A$ …(2)

$A^5 - A^4 = A^3 - A^2 = A - I$ (Using (1))
If n is even $A^n - A^{n-1} = A^2 - A$ …(3)
If n is odd $A^n - A^{n-1} = A - I$ …(4)
Consider n is even
$A^n - A^{n-1} = A^2 - A$ (Using (3))
$A^{n-1} - A^{n-2} = A - I$ (Using (4))
$\Rightarrow A^n - A^{n-2} = A^2 - I$
$\Rightarrow A^n = A^{n-2} + A^2 - I$
$= (A^{n-4} + A^2 - I) + A^2 - I$
$= (A^{n-6} + A^2 - I) + 2(A^2 - I)$
$= (A^2) + \frac{n-2}{2}(A^2 - I)$
$A^n = \left(\frac{n}{2}\right)A^2 - \left(\frac{n-2}{2}\right)I$
$A^{50} = 25A^2 - 24I$
$= 25\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix} - 24\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$
$= \begin{bmatrix}1 & 0 & 0\\ 25 & 1 & 0\\ 25 & 0 & 1\end{bmatrix}$

42. ABCD
Sol. Let f(x) = x
$g(x) = \frac{1}{x}$
f(x) g(x) = 1
periodic
(B) f(x) = x
g(x) = – x
f(x) + g(x) = 0
periodic
(C) f(x) = x
g(x) = x
f(x) – g(x) = 0
periodic
(D) f(x) = $\sin\sqrt{x}$
$g(x) = x^2$
f(g(x)) = sin x
Periodic.

SECTION – B

43. 5
Sol. $\sum_{n=2}^{5}(n-1)(n-\omega)(n-\omega^2) = \sum_{n=2}^{5}n^3 - 1$
$= 220 = 44k \Rightarrow k = 5$

44. 1
Sol. $(1+\tan x^\circ)(1+\tan(x+1)^\circ)...(1+\tan(x+44)^\circ) \geq (1+\tan 1^\circ)(1+\tan 2^\circ)...(1+\tan 45^\circ)$

$(1+\tan 1^o)(1+\tan 2^o)(1+\tan 3^o)...(1+\tan 43^o)(1+\tan 44^o)(1+\tan 45^o)$

$= (1+\tan 1^o)(1+\tan 44^o)(1+\tan 2^o)(1+\tan 43^o)(1+\tan 42^o)...(1+\tan 22^o)(1+\tan 23^0)\times 2$

Using $(1+\tan x^o)(1+\tan(45-x)^o) = 2$

We have

$(1+\tan 1^o)(1+\tan 2^o)(1+\tan 3^o)...(1+\tan 44^o)(1+\tan 45^o) = 2^{23}$

$\Rightarrow (1+\tan x^o)(1+\tan(x+1)^o)(1+\tan(x+2)^o)...(1+\tan(x+44)^o) \le 2^{23}$

Equality occurs at x = 1

45. 1

Sol. There are two cases

Case 1:– $\overset{3}{X}\ \overset{1}{X}\ \overset{1}{X}$ (one pocket has 3 rings)

Number of ways = $3C_1 \times 5C_3 \times 2C_1 = 60$

Case 2 :– $\overset{2}{X}\ \overset{2}{X}\ \overset{1}{X}$ (one pocket has 2 rigs, 2 pockets have 2 each)

Number of ways = $5C_2 \times 3C_2 \times 3C_1 = 90$

Number of total ways = 60 + 90 = 150

Number of ways without restriction = $3^5 = 243$

Required probability = $\frac{150}{243} = \frac{1}{P}$

$\Rightarrow$ [P] = 1

46. 0

Sol. $\frac{1}{(1+x)^2} \le \frac{1}{\sqrt{1+x^3}} \le \frac{1}{1+x}$ for $x \in [2, 3]$

$\Rightarrow \int_2^3 \frac{1}{(1+x)^2}dx \le \int_2^3 \frac{dx}{\sqrt{1+x^3}} \le \int_2^3 \frac{dx}{1+x}$

$\frac{1}{12} \le \int_3^2 \frac{dx}{\sqrt{1+x^3}} \le \ln\frac{4}{3} < 1$

$\Rightarrow \left|\left[\int_2^3 \frac{dx}{\sqrt{1+x^3}}\right]\right| = 0$

47. 3

48. 1

Sol. Coordinates of the point S = $\left(\frac{n^5}{2}, \frac{n^4}{2}, \frac{n}{2}\right)$

$\Rightarrow 2 \times \left(\frac{n^5}{2} + \frac{n^4}{2} + \frac{n}{2}\right) = -1$

$\Rightarrow n(n^4 + n^3 + 1) = -1$

n = – 1 is the only solution.

SECTION – C

49. 00064.00

Sol. $L_1 : \frac{x}{O} = \frac{y-b}{-b} = \frac{z}{c} \Leftrightarrow \vec{r} = b\hat{j} + \lambda\left(-b\hat{j} + c\hat{k}\right)$

$$L_2 : \frac{x-a}{a} = \frac{y}{O} = \frac{z}{c} \Leftrightarrow \vec{r} = a\hat{i} + \mu\left(a\hat{i} + c\hat{k}\right)$$

Equation of plane $P : \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$

Shortest distance between $L_1 \& L_2 = \frac{1}{4}$

50. 10800.00

51. 00004.00

Sol. $f^{-1}(x) = x - 1$

$h^{-1}(x) = \left(gof(x)\right)^{-1}$

$= f^{-1}\left(g^{-1}(x)\right)$

$h^{-1}(x) = x^3 + x$

$\therefore \frac{d}{dx}\left(h^{-1}(x)\right) = 3x^2 + 1$

52. 00000.41

53. 00008.00

Sol. Number of non-negative integral solution of equation 2x + 2y + 3z = 20 are
$^{10+2-1}C_{2-1} + {}^{7+2-1}C_{2-1} + {}^{4+2-1}C_{2-1} + {}^{1+2-1}C_{2-1} = 26$.

54. 00256.00

Sol. We have to de-arrange 5 objects to 6 places
$\therefore$ 44 + 44 + 4[9 + 9 + 3[2 + 2 + 2[1 + 1]]] = 256.